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Thermodynamic re-assessment of the Al–Gd and Gd–Zr systems

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ABSTRACT

Both Al–Gd and Gd–Zr binary systems were re-assessed using the CALPHAD (calculation of phase diagram) method to ensure a multi-component system containing them can be accurately described. Compared with previous assessments, the features of this work are reflected from the following aspects. Firstly, the C15_laves phase Al_2Gd was treated as an intermetallic compound with a negligible homogeneity range using the two-sublattice model $(Al, Gd)_2(Al, Gd)$, meanwhile the liquid phase being described with the substitutional solution model. Secondly, first-principles calculations were carried out to provide theoretical information for optimization. The enthalpies of mixing of $Gd_{1-x}Zr_x$ ($x = 0.25$, 0.5, 0.75) hcp_A3 solid solutions were calculated by applying the Special Quasirandom Structure containing 16 atoms (SQS_16). Finally, the inverse liquid miscibility appearing at high temperatures in previous studies was reduced for the Al–Gd system and eliminated for the Gd–Zr system. The calculated results for both binary systems reproduce the experimental and theoretical data satisfactorily, evidencing the reliability of the thermodynamic descriptions established in this work.

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1. Introduction

Certain rare-earth elements and transition metal elements have been found to be particularly effective in increasing the thermal stability and the strength of aluminum alloys at elevated temperatures since phases that are coherent with the matrix could be formed after addition [1–3]. As it is reported by Mahajan and Kirchoff [4], the Al-4 wt.% Gd-4 wt.% Zr alloy prepared by rapid solidification technique could retain a high micro-hardness until 550 °C. For the design of heat-resistant aluminum alloys with Gd and Zr, knowledge of equilibrium and metastable phase relations is required. Thus, it is useful to build a thermodynamic database for a multi-component Al-based system. Moreover, the Al–Gd–Zr system is a basic constituent of bulk metallic glass systems [5,6]. Their thermodynamic descriptions will facilitate the selection of the composition with optimum glass-forming ability and promote the development of novel materials. Because of the importance of the Al–Gd–Zr system as is shown from the above, both Al–Gd and Gd–Zr binary systems were re-assessed in this work using the CALPHAD method to make some improvements and lay the foundation for the establishment of accurate descriptions of the ternary and multi-component systems containing them.

2. Literature review

2.1. The Al–Gd binary system

The Al–Gd binary phase diagram was first reviewed by Gschneidner and Calderwood [7], and then investigated by Saccone et al. [8] over 0–66.7 at.% Al. Five compounds exist in this system, i.e., Al_3Gd, Al_2Gd, AlGd, Al_2Gd_3, and $AlGd_2$. Colinet et al. [9] measured the enthalpy of formation of all these intermetallic compounds at 298 K by aluminum solution calorimetry, and Sommer and Keita [10] measured that of Al_2Gd, AlGd, and $AlGd_2$. The partial and integral enthalpy of mixing of the liquid phase at 1770 ± 5 K was determined by Kanibolotsky et al. [11] by high-temperature isoperibolic calorimetry.

Thermodynamic optimization of the Al–Gd system has been conducted by Cacciamani et al. [12] and Jin et al. [13]. The work of Cacciamani et al. [12] shows a good fit between the calculated and experimental phase diagram, while the thermodynamic properties of the liquid phase reported later could not be well

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accounted for. Moreover, there was a homogeneity range of the Al_2Gd phase with no experimental evidence. Based on all available experimental data and first-principles calculation results [14], Jin et al. [13] optimized this phase diagram using the modified quasichemical model for the liquid phase and the compound energy formalism for the others. For the sake of model compatibility, the phase diagram of the Al–Gd system was re-assessed in this work, using the substitutional solution model for the liquid phase.

2.2. The Gd–Zr binary system

Copeland et al. [15] have investigated the phase diagram of the Gd–Zr system systematically over the entire composition range by means of differential thermal analysis (DTA), optical metallography (OM), X-ray diffraction (XRD), and electrical resistivity measurements. Only three solution phases liquid, bcc and hcp are reported in this binary system. An eutectic reaction liquid ↔ hcp_Gd + bcc_Zr was detected at 1491 ± 5 K with the eutectic composition being 27.5 ± 1 at.% Zr and a peritectoid reaction bcc_Zr + hcp_Gd ↔ hcp_Zr at 1163 ± 5 K. Subsequently, Copeland and Kato [16] modified the temperature and compositions of liquid and hcp_Gd phases of the eutectic reaction. As yet, no experimental or theoretical thermodynamic properties have been reported. Based on the experimental data on phase diagram, the thermodynamic assessment was first performed by Zinkevich et al. [17]. All the experimental data are well accounted for except that the eutectic composition was calculated as 11.45 at.% Zr which is less than half of the measured one.

Recently, the phase equilibria of the Gd–Zr system were re-investigated by Mattern et al. [18] by electrostatic levitation of the melt and in situ high energy synchrotron X-ray diffraction. The coexistence of bcc_Gd and bcc_Zr phases was detected in a small temperature range, which leads to the occurrence of the eutectoid reaction bcc_Gd ↔ hcp_Gd + bcc_Zr at 1455 ± 10 K instead of the metatectic reaction bcc_Gd ↔ liquid + hcp_Gd at 1488 K calculated by Zinkevich et al. [17]. The liquidus determined by them is different from that by Copeland et al. [15] and the eutectic composition was reported to be 18 ± 2 at.% Zr. Based on the new experimental data of their own, Mattern et al. [18] re-assessed the Gd–Zr phase diagram. All the experimental data are well reproduced. However, there is an inverse miscibility gap of the liquid phase above 2150 K. In addition, the experimental or theoretical thermodynamic properties have not been available up to now, which means the experimental data on phase equilibria are the only data considered in the above assessments. In an attempt to improve the thermodynamic description of this system, the enthalpy of mixing of the hcp_A3 phase was calculated with the first-principles method in this work and then the phase diagram was re-optimized.

3. Thermodynamic model

The substitutional solution model is used to describe all the solution phases including liquid, fcc, bcc, and hcp, with the molar Gibbs energy of pure elements Al, Gd, and Zr in each phase taken from the SGTE data compiled by Dinsdale [19]. All the binary compounds in the Al–Gd system except Al_2Gd were treated as stoichiometric compounds. In the two-sublattice model used for them, Al and Gd occupy the first and the second sublattice, respectively. For a binary C15_laves phase, it is commonly described using the two-sublattice model with two components occupying each sublattice and a ratio of 2:1 for the two sublattices. Thus, the model for the C15_laves phase Al_2Gd was denoted as $(Al, Gd)_2(Al, Gd)_1$. The Gibbs energy per formula unit can be expressed as follows.

$$\begin{aligned} G_{(Al,Gd)_2(Al,Gd)_1} = & \sum_i \sum_j y_i' y_j'' G_{i:j} + 2RT \sum_i y_i' \ln y_i' + RT \sum_i y_i'' \ln y_i'' \\ & + \sum_i \sum_{j>i} y_i' y_j' \sum_k y_k'' \left(\sum_{n=0,1\ldots} {}^{n}L_{i,j:k} \left(y_i' - y_j' \right)^n \right) \\ & + \sum_i \sum_{j>i} y_i'' y_j'' \sum_k y_k' \left(\sum_{n=0,1\ldots} {}^{n}L_{k:i,j} \left(y_i'' - y_j'' \right)^n \right) \\ & + \sum_i \sum_{j>i} \sum_k \sum_{l>k} y_i' y_j' y_k'' y_l'' L_{i,j:k,l} \end{aligned} \quad (1)$$

where i, j, k = Al, Gd, Zr, and l = Gd, Zr. When $i = j$,

$$G_{i:j} = 3^0 G_i + 15000 \quad (2)$$

when $i \neq j$,

$$G_{i:j} = 2^0 G_i + {}^0 G_j + A + BT \quad (3)$$

$$G_{j:i} = 3^0 G_i + 3^0 G_j + 30000 - G_{i:j} \quad (4)$$

In Eqs. (3) and (4), i denotes Al; j denotes Gd; A and B are the parameters to be optimized.

4. First-principles calculation

Based on the first-principles method, the enthalpies of mixing of $Gd_{1-x}Zr_x$ (x = 0.25, 0.5, 0.75) hcp solid solutions at 0 K were calculated by the VASP (Vienna ab initio simulation package) [20,21] software. In this calculation, the SQS_16 proposed by Shin et al. [22] was employed to simulate the hcp solid solutions mentioned above. This structure applies to the calculation of the enthalpy of mixing for hcp solid solution of any two components and its validity has been proved in Cd–Mg, Mg–Zr, and Ti–Zr systems [22]. All-electron projector augmented wave (PAW) function [23,24] proposed by Blöchl was adopted to describe the interaction between ions and electrons and the generalized gradient approximation (GGA) functional by Perdew and Wang (PW91) [25,26] was employed to describe the exchange-correlation energy. Using Monkhorst–Pack method [27], the k-points in the Brillouin zone were produced. The cutoff energy of the plane-wave and the selection of k-points were set as 450 eV and 13 × 13 × 13, respectively, to ensure the convergence of the total energy achieves a precision of 1 meV/atom. During the calculation, every structure is fully relaxed (including volume, atomic sites and shape) until the force between atoms is smaller than 0.01 eV/Å. The potential functions used in this calculation are Al, Gd_3 and Zr_sv. The enthalpies of mixing of $Gd_{1-x}Zr_x$ (x = 0.25, 0.5, and 0.75) hcp solid solutions at 0 K can be calculated by the following formula.

$$\begin{aligned} \Delta H(Gd_{1-x}Zr_x) = & E_{total}(Gd_{1-x}Zr_x) - (1-x)E_{total}(Gd) \\ & - x E_{total}(Zr) \end{aligned} \quad (5)$$

where $E_{total}(Gd_{1-x}Zr_x)$ is the calculated total energy of the $Gd_{1-x}Zr_x$ solid solution, and $E_{total}(Gd)$ and $E_{total}(Zr)$ are the calculated total energies of Gd and Zr in the hcp_A3 structure, respectively.

5. Results and discussion

The phase diagram of the Al–Gd binary system was optimized on the parrot module in the Thermo-calc software developed by Sundman et al. [28], while that of the Gd–Zr system on the PanOptimizer module in the PANDAT 8.1 software [29]. The thermodynamic parameters optimized in this work are summarized in Table 1.

Figs. 1 and 2 present the calculated integral and partial enthalpy of mixing of the liquid phase at 1760 K for the Al–Gd system, respectively, which agree well with the experimental data from

Table 1
Thermodynamic parameters of the Al–Gd and Gd–Zr systems optimized in this work.

Phase	Thermodynamic parameters
Liquid: (Al, Gd)	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Al,Gd}} = -136000 + 25T$
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Al,Gd}} = -50000 + 18T$
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Al,Gd}} = 10000 + 4T$
bcc_A2: $(\mathrm{Al, Gd})_1(\mathrm{Va})_3$	${}^{0}L^{\mathrm{bcc_A2}}_{\mathrm{Al,Gd:Va}} = -52900 + 5T$
Al_3Gd: $(\mathrm{Al})_3(\mathrm{Gd})_1$	$G^{\mathrm{Al_3Gd}}_{\mathrm{Al:Gd}} = -176008 + 32T + 3\,{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}}$
Al_2Gd: $(\mathrm{Al, Gd})_2(\mathrm{Al, Gd})_1$	$G^{\mathrm{Al_2Gd}}_{\mathrm{Al:Gd}} = -153464 + 22.5T + 2\,{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}}$
	$G^{\mathrm{Al_2Gd}}_{\mathrm{Gd:Al}} = 30000 - G^{\mathrm{Al_2Gd}}_{\mathrm{Al:Gd}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + 2\,{}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}}$
	$G^{\mathrm{Al_2Gd}}_{\mathrm{Al,Gd:Gd}} = 91700$
	$G^{\mathrm{Al_2Gd}}_{\mathrm{Al:Al,Gd}} = 31500$
AlGd: $(\mathrm{Al})_1(\mathrm{Gd})_1$	$G^{\mathrm{AlGd}}_{\mathrm{Al:Gd}} = -81000 + 7.9T + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}}$
Al_2Gd_3: $(\mathrm{Al})_2(\mathrm{Gd})_3$	$G^{\mathrm{Al_2Gd_3}}_{\mathrm{Al:Gd}} = -168000 + 12.3T + 2\,{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + 3\,{}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}}$
$AlGd_2$: $(\mathrm{Al})_1(\mathrm{Gd})_2$	$G^{\mathrm{AlGd_2}}_{\mathrm{Al:Gd}} = -86537 + 5.8T + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + 2\,{}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}}$
Liquid: (Gd, Zr)	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Gd,Zr}} = 8289.9 + 9.97T$
	${}^{1}L^{\mathrm{Liq}}_{\mathrm{Gd,Zr}} = 1901.4 - 0.89T$
	${}^{2}L^{\mathrm{Liq}}_{\mathrm{Gd,Zr}} = 9116.3 - 4.05T$
bcc_A2: $(\mathrm{Gd, Zr})_1(\mathrm{Va})_3$	${}^{0}L^{\mathrm{bcc_A2}}_{\mathrm{Gd,Zr:Va}} = 61543.4 - 15.45T$
	${}^{1}L^{\mathrm{bcc_A2}}_{\mathrm{Gd,Zr:Va}} = 27659.3 - 20T$
hcp_A3: $(\mathrm{Gd, Zr})_1(\mathrm{Va})_{0.5}$	${}^{0}L^{\mathrm{hcp_A3}}_{\mathrm{Gd,Zr:Va}} = 32018.6 + 19.34T$
	${}^{1}L^{\mathrm{hcp_A3}}_{\mathrm{Gd,Zr:Va}} = 6623.6 - 6T$
	${}^{2}L^{\mathrm{hcp_A3}}_{\mathrm{Gd,Zr:Va}} = -15.36T$

Kanibolotsky et al. [11]. The calculated results of Cacciamani et al. [12] are also shown for comparison. For the measured enthalpy of formation of Al–Gd binary compounds at 298 K, data reported by Colinet et al. [9] and Sommer and Keita [10] are basically the same except that of $AlGd_2$, in which the difference is as large as 5700 J/mol. Since the enthalpy of formation of $AlGd_2$ measured by Colinet et al. [9] is more close to the first-principles calculation results from the literature, that by Sommer and Keita [10] was given a low weight during the optimization. Fig. 3 presents the calculated enthalpy of formation of the compounds at 298 K. It has a better agreement with the experimental data and first-principles calculation results compared with the assessment result of Cacciamani et al. [12]. This shows very well in the AlGd, Al_2Gd_3, and $AlGd_2$ compounds. The calculated phase diagram of the Al–Gd system is shown in Fig. 4. As can be seen from it, there is almost no homogeneity range for the Al_2Gd phase. The temperature of the peritectic reaction liquid + AlGd ↔ Al_2Gd_3 has been lowered, which minimizes the difference between the calculated and experimental temperature. It should be noted that a large range of inverse liquid miscibility emerges above 3868 K in the Al–Gd phase diagram

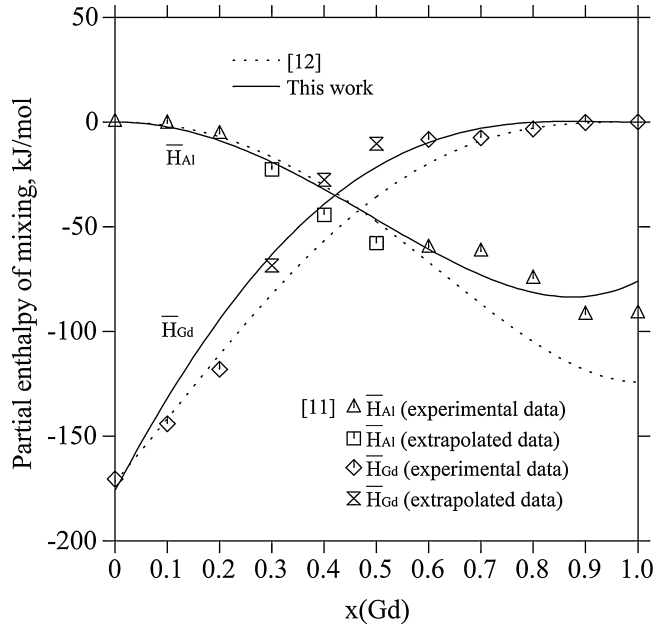

Fig. 2. Calculated partial enthalpy of mixing of the liquid phase in the Al–Gd system at 1760 K.

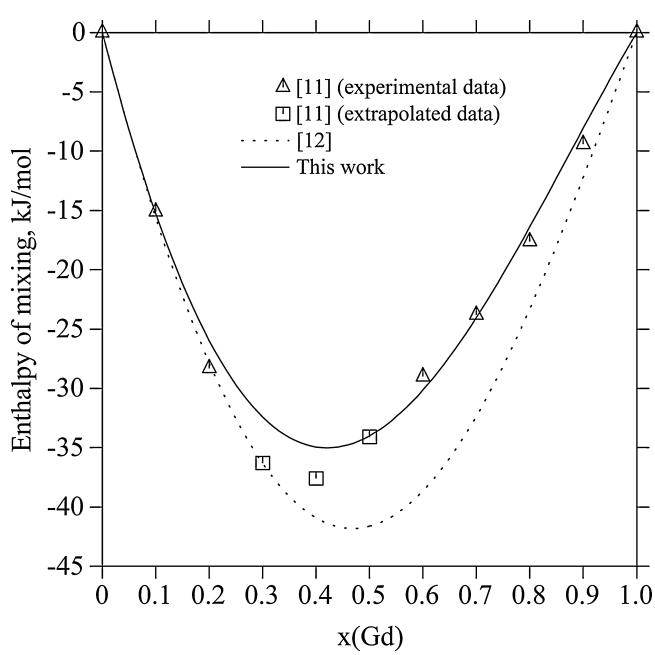

Fig. 1. Calculated enthalpy of mixing of the liquid phase in the Al–Gd system at 1760 K.

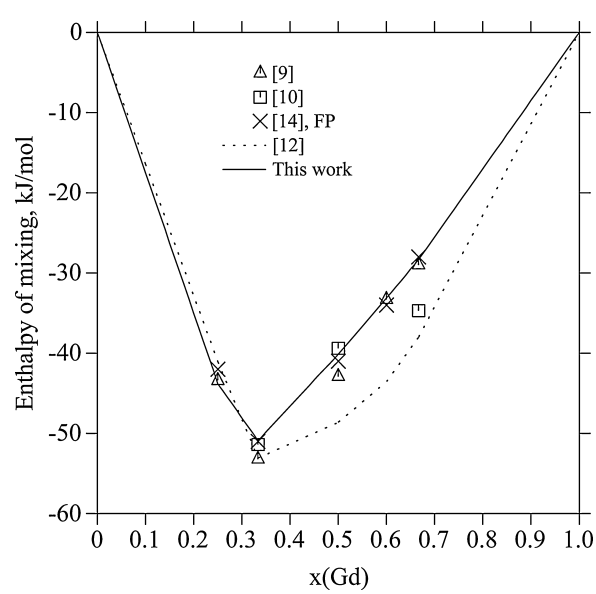

Fig. 3. Calculated enthalpy of formation of Al–Gd binary compounds at 298 K (FP denotes first-principles calculation, the same hereinafter).

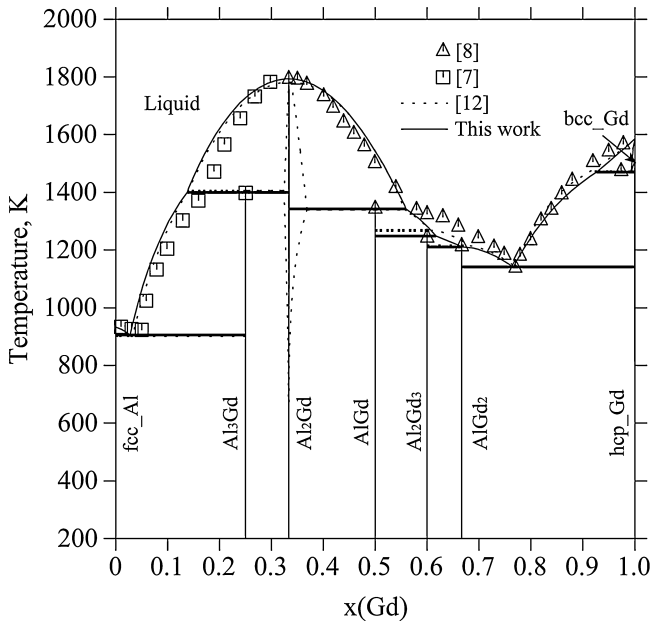

Fig. 4. Calculated phase diagram of the Al–Gd system.

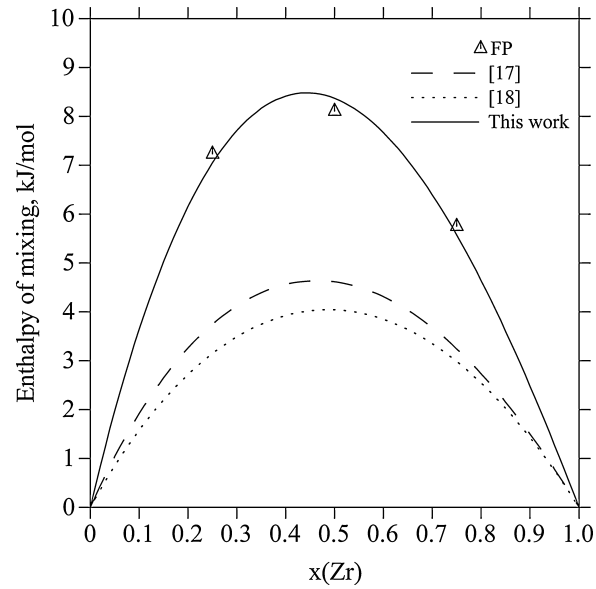

Fig. 5. Calculated enthalpy of mixing of the hcp_A3 solid solution in the Gd–Zr system at 298 K.

calculated by Cacciamani et al. [12], which is not reasonable from a thermodynamic point of view and should be avoided. In this work, only a small inverse liquid miscibility exists above 5812 K that is about 4000 K higher than the maximum liquidus temperature. Thus, it will not exert an influence on the thermodynamic description of any multi-component system containing the Al–Gd binary system. Comparing this work and the work of Jin et al. [13], it could be noticed that thermodynamic properties of the liquid phase are well accounted for by both the substitutional solution model and the modified quasichemical model, yet there is still a little difference between the two sets of calculation results. Considering the short-range ordering, the enthalpy of mixing of the liquid phase calculated by Jin et al. [13] shows a minimum value around $x_{Gd} = 1/3$, while in this work the minimum value lies around $x_{Gd} = 0.4$. The calculated partial enthalpy of mixing by Jin et al. [13] varies slowly at first with the addition of a second component and changes quickly around $x_{Gd} = 1/3$, which is not the case in this work. Because of these differences, the liquidus calculated by the modified quasichemical model is non-symmetric around the Al_2Gd phase and the liquid composition in the

Table 2
Calculated invariant reactions in the Al–Gd system.

Reaction	T (K)	Phase composition x_{Gd}			Ref.
Liquid ↔ fcc_Al + Al_3Gd	923	0.050			[7,8]
	903	0.032	0	0.250	[12]
	907				[13]
	906	0.0274	0	0.250	This work
Liquid + Al_2Gd ↔ Al_3Gd	1398	0.160			[7,8]
	1403	0.140	0.324	0.250	[12]
	1398				[13]
	1400	0.137	0.333	0.250	This work
Liquid ↔ Al_2Gd	1793	0.333			[7,8]
	1782	0.333	0.333		[12]
	1800				[13]
	1793	0.333	0.333		This work
Liquid + Al_2Gd ↔ AlGd	1343	0.560			[7,8]
	1341	0.557	0.368	0.500	[12]
	1325				[13]
	1343	0.559	0.333	0.500	This work
Liquid + AlGd ↔ Al_2Gd_3	1243	0.690			[7,8]
	1268	0.606	0.500	0.600	[12]
	1213				[13]
	1248	0.617	0.500	0.600	This work
Liquid + Al_2Gd_3 ↔ $AlGd_2$	1213	0.720			[7,8]
	1214	0.683	0.600	0.667	[12]
	1187				[13]
	1211	0.6674	0.600	0.667	This work
Liquid ↔ $AlGd_2$ + hcp_Gd	1138	0.770			[7,8]
	1141	0.770	0.667	0.9999	[12]
	1141				[13]
	1141	0.766	0.667	0.9998	This work
bcc_Gd ↔ Liquid + hcp_Gd	1473		0.905		[7,8]
	1471	0.993	0.913	0.9996	[12]
	1474				[13]
	1471	0.993	0.926	0.9998	This work

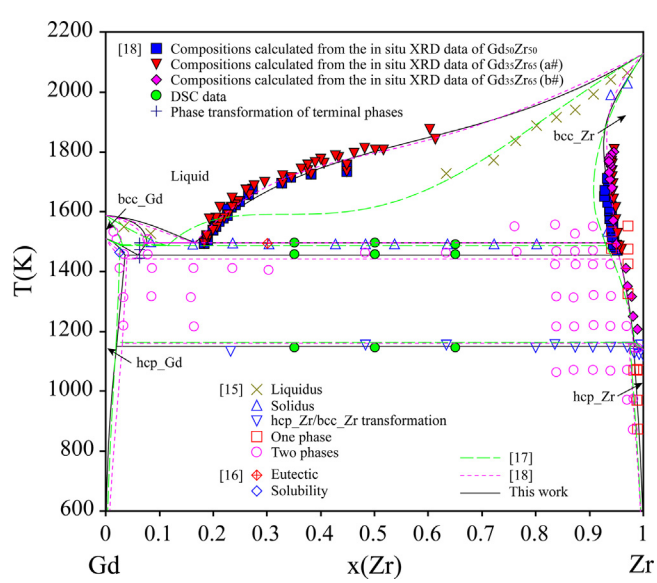

Fig. 6. Calculated phase diagram of the Gd–Zr system.

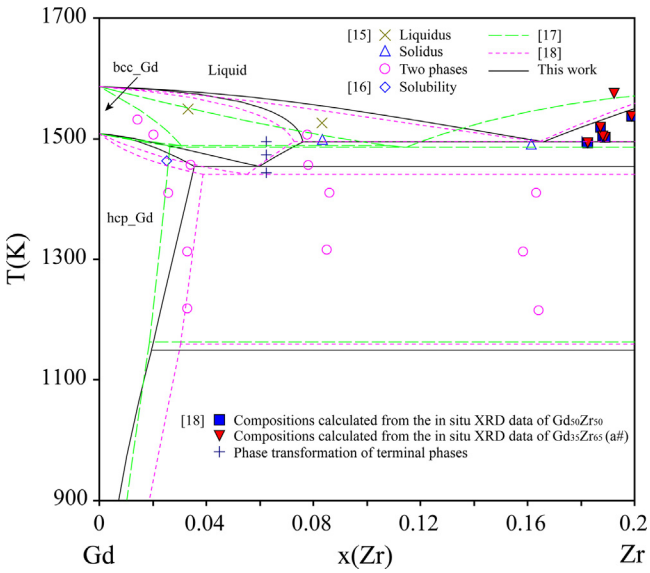

Fig. 7. Gd-rich region of the calculated Gd–Zr phase diagram.

peritectic reaction liquid + $Al_2Gd \rightarrow Al_3Gd$ is overestimated by about 4 at.% Gd, while the liquidus calculated by the substitutional solution model seems symmetric around Al_2Gd, which leads to an underestimation of that composition by about 2 at.% Gd. In general, the calculated temperatures of invariant reactions by the present work have a better agreement with the experimental data than previous assessments [12,13], as can be seen from Table 2.

As is shown in Fig. 5, the enthalpy of mixing for the hcp_A3 solid solution of the Gd–Zr system obtained from thermodynamic calculation agrees very well with that from first-principles calculation. However, it is not the same case for the calculation results of Zinkevich et al. [17] and Mattern et al. [18], both of which are almost half of the first-principles results. Fig. 6 shows the calculated Gd–Zr phase diagram, compared with the experimental data [15,16,18] and the phase diagrams optimized by Zinkevich et al. [17] and Mattern et al. [18]. During the optimization, the experimental data of Mattern et al. [18] were mainly considered, and the results of Copeland et al. [15,16] were considered when optimizing the solvus of hcp_Gd and hcp_Zr phases as well as the solidus of bcc_Zr phase. The phase diagram calculated by this work is similar to that by Mattern et al. [18]. Taking them for comparison, the calculated temperatures of invariant reactions agree with the experimental data a little better in this work. The Gd-rich region and Zr-rich region are enlarged, as shown in Fig. 7 and Fig. 8, respectively. From these figures, it can be seen that individual experimental data for the hcp_Gd + bcc_Zr and hcp_Zr phase fields [15] could not be explained. Since the difference between calculated and experimental results is less than 1% which is within the experimental error, the present calculation results are acceptable. In addition, the inverse liquid miscibility above 2150 K in the Gd–Zr phase diagram calculated by Mattern et al. [18] is completely eliminated in this work, which is an important improvement for the thermodynamic description of this binary system. The calculated invariant reactions and compositions of equilibrium phases are listed in Table 3.

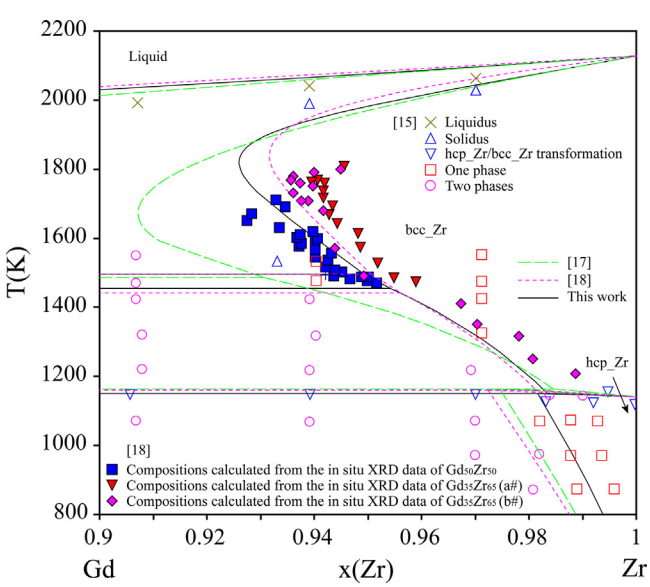

Fig. 8. Zr-rich region of the calculated Gd–Zr phase diagram.

Table 3
Calculated invariant reactions in the Gd–Zr system.

Reaction	T (K)	Phase composition x_{Zr}			Ref.
Liquid ↔ bcc_Gd + bcc_Zr	1495 ± 10				[18]
	1495	0.164	0.073	0.951	[18]
	1495	0.166	0.076	0.949	This work
bcc_Gd ↔ hcp_Gd + bcc_Zr	1455 ± 10				[18]
	1441	0.055	0.038	0.956	[18]
	1454	0.060	0.035	0.954	This work
bcc_Zr + hcp_Gd ↔ hcp_Zr	1145 ± 10				[18]
	1159	0.981	0.030	0.972	[18]
	1149	0.983	0.019	0.982	This work

6. Conclusions

Based on the experimental information on phase diagram and thermodynamic properties and the first-principles calculation results, the Al–Gd binary phase diagram was re-assessed, in which the Al_2Gd phase has almost no homogeneity range. The calculated results reproduce the experimental data very well. The inverse liquid miscibility in the previous assessment was reduced a lot and only appears above 5812 K, which ensures the validity of the thermodynamic description of the multi-component system containing Al–Gd at high temperatures. As for the Gd–Zr system, the enthalpies of mixing of $Gd_{1-x}Zr_x$ (x = 0.25, 0.5, 0.75) hcp_A3 solid solutions at 0 K were calculated by the first-principles method. Based on these results and available experimental data, the phase diagram of the Gd–Zr system was re-assessed. The inverse liquid miscibility in the previously assessed phase diagram was completely eliminated and good fit was found between the calculated results and experimental and theoretical data. It should be noted that though first-principles calculations were carried out to provide partial theoretical information for the assessment, experimental investigation on the thermodynamic properties of the Gd–Zr system is still needed to further improve the current thermodynamic description.

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